Sharp Counts and Small Integer Witnesses for Tight Cone Rules

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Abstract

The cone rule for right-hand-side sensitivity in linear programming generates an inner approximation of a simplicial feasibility cone from individual coordinate ranges. We count the nonzero right-hand-side directions at which this approximation is exact, including boundary points and coordinate extreme rays. Let q be the number of columns of the inverse basis matrix having at least two nonzero entries. If $q = 3$, there are at most three tight directions; if $q \geq 4$, there is at most one. Both bounds are attained. The proof separates supports of size one, two, and at least three, and uses an increasing-ratio cycle contradiction to establish uniqueness across different assignments of columns to missing axes. For an integral inverse basis matrix with entries of absolute value at most M, every tight direction when $q \geq 3$ has a primitive nonnegative integer representative of height at most M^2. An explicit three-dimensional family shows that quadratic growth is necessary. A finite exact candidate procedure includes zero endpoints and recession generators without genericity assumptions.

Contribution and significance

Fampa and Lee introduced the cone rule and characterized its interior tightness when the feasibility cone has no coordinate extreme ray. Their characterization gives uniqueness for a fixed assignment and a polynomial test. The present contribution is a sharp count across all assignments and all nonzero feasible supports, while allowing coordinate rays. The distinction matters: zeros change which finite endpoints and recession directions can generate a missing extreme ray, and interior arguments alone do not control the boundary.

The controlling parameter is the number of axes not already supplied by pure inverse-basis columns, rather than ambient dimension. Large-support witnesses exclude every competing support, whereas sparse witnesses are counted by disjoint requirements on two- and three-entry columns. This yields the exceptional three-direction configuration and the one-direction bound beyond it. The integer-height bound supplies a separate certificate-size consequence; it concerns the inverse basis, not an unspecified bound on the original basis. These are structural results about exactness, not improved performance guarantees for a linear-programming solver.

1. Definitions and statements

Fix an invertible $G \in \mathbb{R}^{m\times m}$, $m \geq 3$, and write $H = G^{-1} = [h_1|\cdots|h_m]$. The feasibility cone is $G\mathbb{R}^m_+$. A nonzero feasible right-hand side b has coordinates $y = Hb \geq 0$. We do not require $b \geq 0$. For each column set

$$L_i = \max_{h_{ki}>0} \frac{-y_k}{h_{ki}}, \qquad U_i = \min_{h_{ki}<0} \frac{-y_k}{h_{ki}},$$

with empty extrema $-\infty, +\infty$. Thus $y + th_i \geq 0$ exactly for $t \in [L_i, U_i]$. Define the two generators

$$g_i^- = \begin{cases} y + L_i h_i & L_i > -\infty, \\ -h_i & L_i = -\infty, \end{cases} \qquad g_i^+ = \begin{cases} y + U_i h_i & U_i < \infty, \\ h_i & U_i = \infty. \end{cases}$$

All belong to $\mathbb{R}^m_+$. The transformed cone rule is

$$C_H(y) = \operatorname{cone}\{g_i^-, g_i^+ : 1 \leq i \leq m\}.$$

This is the inverse-basis form of the rule in [1]. We call y *tight* if $C_H(y) = \mathbb{R}^m_+$, identifying positive scalar multiples as a single direction. In particular, a zero endpoint is not a generator of an axis.

A column is *pure* if its support has size one. Their supporting rows are distinct, since H is invertible. Let C be those rows, $R = [m] \setminus C$ the missing rows, and I the nonpure columns. Then $|R| = |I| =: q$.

Theorem 1.1. *For every real invertible H, the number of tight directions is at most three when $q = 3$, and at most one when $q \geq 4$. Both bounds are sharp in every ambient dimension admitting the indicated q. When $q = 3$, three directions occur precisely when the nonpure columns are supported on the three edges of a triangle on R, each with opposite signs at its two entries. The tight directions in this case are the three coordinate axes indexed by R.*

Theorem 1.2. *Suppose H is integral, invertible, $q \geq 3$, and $M = \max_{ki} |H_{ki}|$. Every tight direction has a primitive integer representative $v \geq 0$ with $\|v\|_\infty \leq M^2$. Singleton and two-coordinate supports admit the stronger bounds 1 and M, respectively. For every odd $M \geq 3$ there is a 3×3 example whose unique tight direction has primitive height $M(M-1)$.*

The restriction $q \geq 3$ is essential for finiteness: for $H = I$ every positive y is tight. The proofs below do not claim a classification for $q \leq 2$.

2. Axis generators and large supports

Lemma 2.1. *A nonzero $y \geq 0$ is tight exactly when every coordinate axis contains a positive generator $g_i^\pm$. A pure column supplies its supporting axis by a recession generator. If $|\operatorname{supp} y| \geq 3$, each nonpure column can supply at most one missing axis, and tightness induces a bijection $i : R \to I$ with*

$$y + \theta_j h_{i(j)} = \mu_j e_j, \qquad \theta_j \neq 0, \quad \mu_j > 0. \tag{2.1}$$

Proof. A nonnegative linear combination equals a positive multiple of e_j only if one of its nonzero summands lies on that axis. This proves the first assertion. A pure column has an infinite feasible direction yielding its supporting axis. For a large-support y, its finite endpoint cannot lie on any axis, because altering one coordinate leaves at least two positive coordinates unchanged. A nonpure recession vector is not an axis vector. Finally, if $y + \theta h_i = \mu e_j$ and $y + \phi h_i = \nu e_k$ for $j \neq k$, subtracting gives $\operatorname{supp} h_i \subseteq \{j, k\}$ and then $\operatorname{supp} y \subseteq \{j, k\}$, a contradiction. There are q missing axes and exactly q available nonpure columns, giving the bijection and nonzero parameters. □

Lemma 2.2. *Assume $q \geq 3$. There is at most one tight direction with support of size at least three, without fixing its column-to-axis assignment.*

Proof. Let y, z be two such witnesses. Index the columns by their targets $j \in R$ for y. Let π be the permutation of R giving their targets for z. Equations (2.1) yield

$$z - \lambda_j y = \nu_{\pi(j)} e_{\pi(j)} - \lambda_j \mu_j e_j, \qquad \lambda_j = \phi_j / \theta_j. \tag{2.2}$$

Choose $a \in \operatorname{supp} y \setminus \{j, \pi(j)\}$. Then $z_a = \lambda_j y_a$ implies $\lambda_j \geq 0$. It is nonzero, since both line parameters are nonzero; hence $\lambda_j > 0$. If $\pi(j) \neq j$, the j coordinate gives $y_j > 0$ and $z_j/y_j < \lambda_j$. Along a nontrivial permutation cycle, each coordinate is positive in y, and the $\pi(j)$ coordinate gives

$$z_j/y_j < \lambda_j < z_{\pi(j)}/y_{\pi(j)}.$$

Strict increase around the cycle is impossible, so π is the identity. For any two distinct $j, k \in R$ choose a positive coordinate of y outside $\{j, k\}$. It follows from (2.2) that $\lambda_j = \lambda_k$. For each coordinate a, take $j \neq a$ in R; then $z_a = \lambda_j y_a$. Thus z is a positive multiple of y. □

3. Sparse supports and incompatibility

Lemma 3.1. *If y is tight with support $\{a\}$, each $j \in R \setminus \{a\}$ requires a distinct nonpure column supported exactly on $\{a, j\}$ with opposite entry signs. These columns, together with the pure columns, are sufficient for tightness.*

If y is tight with support $S = \{a, b\}$, each $j \in R \setminus S$ requires a distinct column supported exactly on $S \cup \{j\}$, with its a, b entries of the same sign, its j entry of the opposite sign, and

$$|h_{ai}| : |h_{bi}| = y_a : y_b.$$

Such a column determines the pair direction and its missing target uniquely. If $S \subseteq R$, supplying the active axes also requires columns supported within S.

Proof. For a target outside $\operatorname{supp} y$, an axis equation $y + \theta h_i = \mu e_j$ forces the asserted support, signs, and ratios. The line parameter is nonzero. Conversely such an equation is a feasible endpoint: any cancelled positive coordinate places it on the corresponding finite range bound. In the singleton case the active axis is already supplied by a zero-parameter endpoint, or by a pure recession generator. Indeed, if any column has a nonzero entry outside a, one of its finite bounds is zero; invertibility ensures such a column for $m \geq 2$. This proves sufficiency. For pair support, a column reaching a or b must have support within S; a pure column can do so only when its row is in $S \cap C$. Thus when $S \subseteq R$ the needed columns are nonpure. A three-entry column with a singleton minority sign identifies its minority target, the complementary pair, and its ratio uniquely. □

Lemma 3.2. *When $q \geq 3$, a large-support tight direction cannot coexist with a singleton or pair tight direction.*

Proof. Every nonpure column must participate in the large-support bijection. A mixed-sign two-entry column cannot participate: off its target the equation (2.1) would either force at most two coordinates of y positive or require two opposite signs to be cancelled by one scalar. A singleton requires at least $q - 1$ such columns and is therefore incompatible.

For a pair S, a required three-entry column can serve a large-support witness only at its minority target j. Its two other possible targets leave opposite off-target signs. Hence the large support is contained in $S \cup \{j\}$. If there are two missing targets outside S, intersecting these containments forces support inside S, a contradiction. The only remaining case is $q = 3$ and $S \subseteq R$. A column within S with mixed signs is excluded as above. A nonpure same-sign column within S can serve the large witness only at the sole row of $R \setminus S$. To supply the two active axes for the pair, either one mixed-sign column or at least two same-sign columns within S are necessary: one same-sign column cannot hit both positive axes, since subtracting the two endpoint equations would give opposite signs. The latter two columns cannot both be assigned to the same missing row in a bijection. This is the remaining contradiction. □

4. Sharp counting

Proof of Theorem 1.1. The preceding lemmas settle the case of a large-support witness. Suppose all witnesses have support at most two.

Two distinct singleton witnesses need at least $2q-3$ distinct two-entry columns: their required stars share at most the edge joining their centers. For $q \geq 4$ this exceeds q. A singleton and a pair need at least $(q-1)+(q-2) > q$ columns of disjoint support sizes. Two different pair directions need disjoint sets of at least $q-2$ three-entry columns. For $q \geq 5$ this again exceeds q. In the remaining $q = 4$ case, equality would force both pair supports to lie in R and all four nonpure columns to have three entries. There would then be no within-pair column to supply their active axes. This proves the bound one.

Let $q = 3$. Two singleton witnesses require exactly the three edges of a triangle on R. To see the qualification on the centers, a center outside R needs all three columns, whereas a second star shares at most one; the total would exceed three. Thus both centers lie in R, and their two stars exhaust the three triangle edges. All are mixed-sign, and by Lemma 3.1 the third singleton is tight as well; no pair can occur. If exactly one singleton occurs, it consumes at least two two-entry columns, leaving room for at most one pair direction. If no singleton occurs, three pair directions would require three distinct three-entry columns, each pair support in R, leaving no column within a pair to supply its active axes. Thus there are at most two pairs. This proves the count and the equality characterization.

For sharpness at $q = 3$, take

$$H = \begin{pmatrix} 1 & 0 & -\gamma \\ -\alpha & 1 & 0 \\ 0 & -\beta & 1 \end{pmatrix}, \qquad \alpha, \beta, \gamma > 0, \quad \alpha\beta\gamma \neq 1.$$

Its determinant is $1 - \alpha\beta\gamma$, and its three axes are tight. For $q \geq 4$, take $H = (q+1)I - \mathbf{1}\mathbf{1}^T$. It is invertible, and $y = \mathbf{1}$ is tight since $y + h_j = (q+1)e_j$. Append identity blocks for larger ambient dimension; use witnesses zero on those added coordinates. Pure columns supply the added axes, so both constructions retain the claimed counts. □

5. Integer height and exact recognition

Proof of Theorem 1.2. Singletons have height one. For a pair, $R \setminus \operatorname{supp} y$ is nonempty because $q \geq 3$. A required three-entry column determines the two positive coordinates as the absolute values of its corresponding entries, followed by division by their gcd; hence height at most M.

For large support choose two distinct missing targets j, k and their distinct columns h_i, h_l. Choose $a \in \operatorname{supp} y \setminus \{j, k\}$. Define an integer vector by

$$v_s = h_{al} h_{si} \ (s \neq j), \qquad v_j = h_{ai} h_{jl}.$$

The witness equations give $h_{al} h_{si} = y_a y_s / (\theta_k \theta_j)$ for $s \neq j$ and the same identity for $s = j$. Thus v is a nonzero scalar multiple of y. Change its global sign and divide by the gcd to get a nonnegative primitive representative. Every entry is a product of two entries of H, proving the upper bound.

For odd $M \geq 3$ put $a = M, b = M-1, c = M-2$ and

$$H = \begin{pmatrix} 1 & -a & -b \\ -a & 1 & -c \\ -b & -c & 1 \end{pmatrix}, \qquad v = (ab, ac, bc)^T.$$

The determinant $1 - a^2 - b^2 - c^2 - 2abc$ is negative. Moreover $v + ch_1$, $v + bh_2$, and $v + ah_3$ are positive multiples of the three axes, so v is tight. The integers a, b, c are pairwise coprime, since

M is odd; consequently $\gcd(ab, ac, bc) = 1$. Its height is $ab = M(M-1)$. By Theorem 1.1 and Lemma 3.2, its large-support witness is the only tight direction. □

Proposition 5.1. *For rational invertible H with $q \geq 3$, all tight directions can be found by a finite exact procedure polynomial in dimension and input bit length.*

Proof. List the coordinate axes. List pair candidates from each three-entry column with a unique minority sign, using the other two absolute entries as the positive pair. These include every small-support witness by Lemma 3.1. Fix distinct $j, k \in R$. For every ordered pair of distinct nonpure columns h_i, h_l, compute

$$\operatorname{span}(h_i, e_j) \cap \operatorname{span}(h_l, e_k).$$

Both planes have dimension two. If the intersection is a line, keep its nonnegative orientation if one exists. If the planes coincide, their common plane contains both e_j, e_k and hence is their coordinate plane; it contains no support of size three. If the intersection is zero, discard it. Lemma 2.1 ensures that every large-support witness occurs in this list. Finally evaluate all actual finite endpoints and recession vectors exactly for every candidate and retain precisely those supplying every axis. This test rejects zero endpoints. Deduplicate positive proportional vectors. There are polynomially many candidates, and only rational linear algebra of bounded matrix dimension per pair and polynomially many arithmetic operations are used; standard rational arithmetic has polynomial bit complexity here. □

Independent nonzero column scalings do not change a feasible line, its endpoint set, or its recession rays. Therefore one may clear denominators separately in each rational column before using the height bound, but M then means the resulting integer height, not the original numerator bound.

6. Verification and limitations

The proofs of the counting and height theorems are symbolic and apply to every stated real or integer input. The companion exact checker evaluates range endpoints using rational arithmetic, recognizes positive coordinate rays, and enumerates the finite candidate list. Its examples include the three-ray triangle, large-support witnesses, mixed boundary cases, pure-column padding, and the sharp height family. Negative controls reject smaller counting bounds, a linear height bound, and a singular triangle. These finite tests help detect implementation errors; they do not replace the support and cycle proofs.

The result does not estimate the volume fraction captured when tightness fails, classify $q \leq 2$, or prove a complexity advantage for an LP implementation. The closest-source comparison is to the explicit interior theorem in [1]; it is not a claim to have exhausted all priority evidence.

Manuscript information

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References

[1] M. Fampa and J. Lee, *On the 100% Rule of Linear Programming*, arXiv:2610.12665v1, 2026, Sections 2–3, Theorem 5.6, and Section 11. https://arxiv.org/abs/2610.12665v1.